

Sensitivity, Informativeness, and Misspecification in GMM Estimation

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Motivation

- Sensitivity to model specification is central to assessing the **robustness** of empirical conclusions.
- Reduced form: a rich, standardized toolkit to check omitted variable bias formula. ([Rosenbaum and Rubin 1983](#); [Imbens 2003](#); [Oster 2019](#); [Cinelli and Hazlett 2020](#))
- Structural estimators: [far less standardized](#).
 - the map from moment conditions to parameters is implicit and nonlinear;
 - misspecification is prevalent in applied work.

Sensitivity is a local robustness diagnostic

- **Sensitivity** (Andrews, Gentzkow, and Shapiro 2017, AGS): how $\hat{\theta}$ responds to each estimation moment.
- A **local** deviation of the moments, evaluated at **correct specification**.
- But in their analysis of Berry, Levinsohn, and Pakes (1995) (BLP):
 - the J -test rejects strongly
 - the moments are **globally (fixed) misspecified**.

Sensitivity diagnostics meet misspecification

- The J -test rejects in practice:
 - Berry, Levinsohn, and Pakes (1995) (BLP): $J = 345$ (df 25).
 - Blundell, Pistaferri, and Preston (2008) (BPP): $J = 539$ (df 290).
- Misspecification $\Rightarrow \hat{\theta}$ converges to the pseudo-true value θ_0 , the unique minimizer of the GMM criterion, with

$$g = \mathbb{E}[g(X_i, \theta_0)] \neq 0.$$

- **Extend AGS:** sensitivity as a local deviation around the fixed misspecified θ_0

Setup

- Sample moments $\hat{g}(\theta)$.
- Joint asymptotic distribution of the estimator and the moments:

$$\sqrt{n} \begin{pmatrix} \hat{\theta} - \theta_0 \\ \hat{g}(\theta_0) - g \end{pmatrix} \rightsquigarrow \begin{pmatrix} \tilde{\theta} \\ \tilde{g} \end{pmatrix} \sim \mathcal{N} \left(0, \begin{pmatrix} \sigma_{\theta\theta} & \sigma_{\theta g} \\ \sigma_{g\theta} & \sigma_{gg} \end{pmatrix} \right).$$

DEFINITION 1 (*misspecification-robust sensitivity, MRS*)

$$\Lambda = \sigma_{\theta g} \sigma_{gg}^{-1} \quad (p \times q).$$

In the limit, $\mathbb{E}[\tilde{\theta} \mid \tilde{g}] = \Lambda \tilde{g}$: the regression coefficient of the estimator on the moments.

Correct specification: $\Lambda_{AGS} = -(G'WG)^{-1}G'W$.

Sensitivity requires the influence function

- Asymptotically linear representation:

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi(X_i) + o_p(1), \quad \nu_i = g(X_i, \theta_0) - g.$$

- Then $\Lambda = \mathbb{E}[\psi \nu'] \mathbb{E}[\nu \nu']^{-1}$: we need $\psi(X_i)$ at the pseudo-true value.
- Correct specification: $\psi_i = \Lambda_{AGS} \nu_i$.
- Misspecification ($g \neq 0$): sampling variation in the Jacobian $\hat{G}$ and the estimated weight $\hat{W}(\hat{\phi})$, with $\hat{\phi}$ a first-step estimator, enters ψ at first order, through influence functions $\gamma_i, \omega_i, \psi^\phi$.

Proposition 1 derives ψ and these channels.

Channel decomposition

PROPOSITION 1 (*decomposition of GMM influence functions*)

One-step GMM with estimated weight $\widehat{W}$:

$$\psi^{1s,W}(X_i) = -A^{-1} \left[\underbrace{G'W(g_i - g)}_{\text{moment}} + \underbrace{(g'W \otimes I_p) \gamma_i}_{\text{Jacobian}} + \underbrace{(g' \otimes G') \omega_i}_{\text{weight matrix}} \right].$$

Two-step adds $B\psi^\phi(X_i)$ inside the bracket; iterated replaces A^{-1} by $(A + B)^{-1}$.

DEFINITION 2 (*informativeness*)

$$\Delta_k = \frac{\sigma_{\theta_k g} \sigma_{gg}^{-1} \sigma_{g\theta_k}}{\sigma_{\theta_k \theta_k}} = R^2 \text{ of } \hat{\theta}_k \text{ on the moments } \in [0, 1].$$

Informativeness, and when $\Delta_k < 1$

- We interpret informativeness as **structural efficiency**: the proportion of the asymptotic variance explained by the moments.
- Extends Andrews, Gentzkow, and Shapiro (2020)
 - they derive the informativeness of **descriptive statistics** for correctly specified GMM estimator
 - replacing **descriptive statistics** by g , $\Delta = 1$ under correct specification
- $\Delta_k < 1$ when the Jacobian channel, the weight-matrix channel, or the preliminary-estimator channel matter.
- Δ_k vs. the J -test.

Optimal minimum distance

(Proposition 2 in the paper)

- Minimum distance, $g(X_i, \theta) = m_i - f(\theta)$, with the optimal weight $\widehat{W} = \widehat{V}^{-1}$.
- Jacobian is nonrandom and there is no first step.
- Only the **weight-matrix channel** survives: we show

$$\psi_i = M_\nu \nu_i (1 - \nu_i' W g).$$

- Under Gaussian moments, we show:

$$\Delta_k = \frac{1}{1 + g' W g}.$$

- The **optimal weight buys efficiency at the price of informativeness.**

The weighting ladder

$V = I_q$, Gaussian, common misspecification $g_k = \delta$. Same population weight; differ in how much is estimated:

$$\underbrace{\Delta_k^{OMD} = \frac{1}{1 + q\delta^2}}_{\text{optimal weight}} \quad \underbrace{\Delta_k^{DWMD} = \frac{1}{1 + 2\delta^2}}_{\text{diagonal}} \quad \underbrace{\Delta_k^{EWMD} = 1}_{\text{identity}}.$$

- OMD channel grows with q ; in BPP, $q = 325$.
- Asymptotic counterpart of Altonji and Segal (1996).

Sensitivity to the first step, and iteration

PROPOSITION 3 (*sensitivity to the first-step estimator*)

$$\Lambda_{\phi} = \text{plim} \frac{\partial \hat{\theta}(\hat{\phi})}{\partial \hat{\phi}'} = -A^{-1}B.$$

Iterated GMM as a fixed point

PROPOSITION 4 (*geometric convergence of iterated GMM*)

For every $\tilde{\rho} \in (\rho(-A^{-1}B), 1)$, there is $C_{\tilde{\rho}}$ with

$$\|\psi^{ss}(X_i) - \psi^{it}(X_i)\|_{L^2(P)} \leq C_{\tilde{\rho}} \tilde{\rho}^{s-1}.$$

Hence $\text{Var}(\psi^{ss}) \rightarrow \text{Var}(\psi^{it})$ and $\Delta_k^{ss} \rightarrow \Delta_k^{it}$ at the same rate.

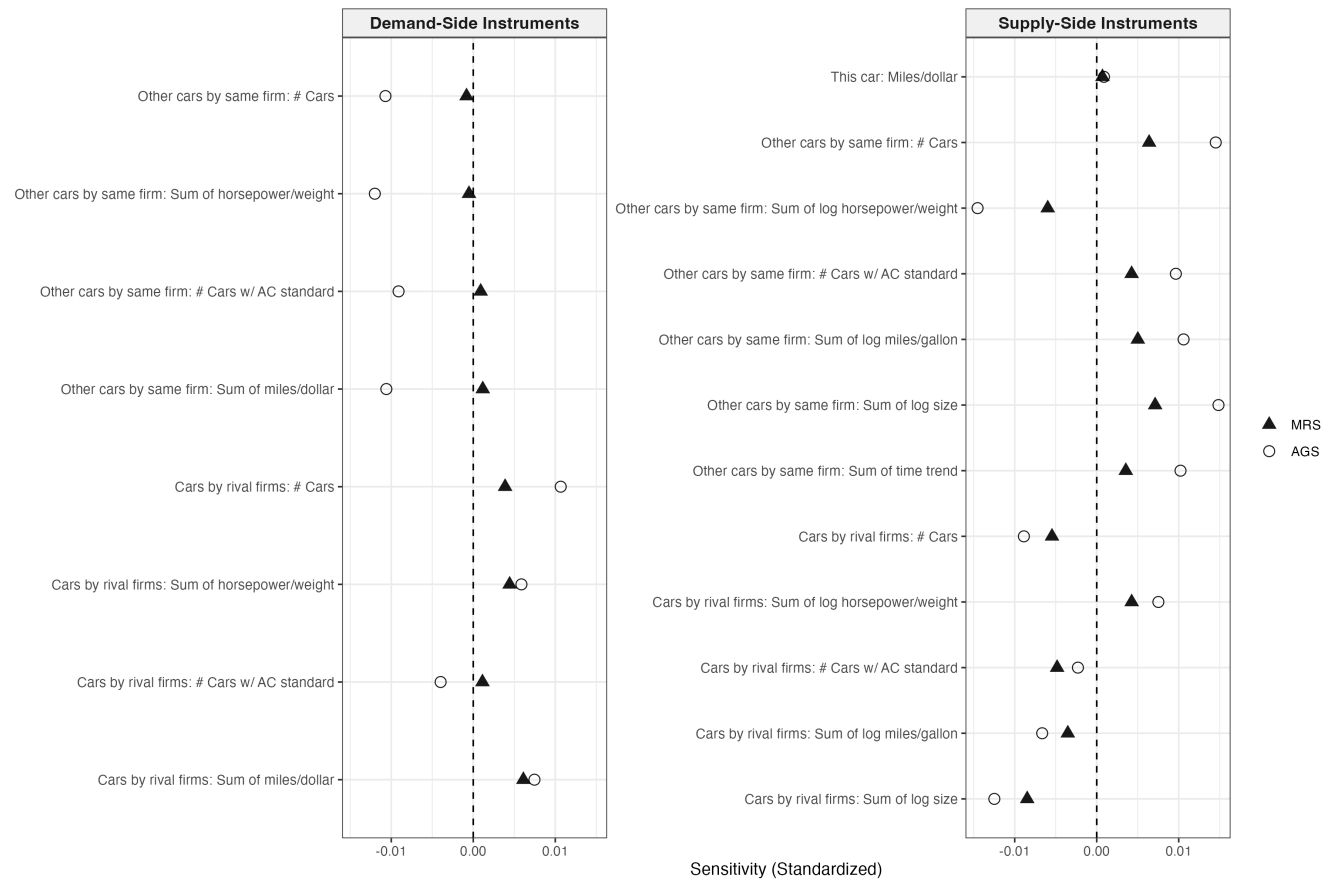
- Hansen and Lee (2021) show convergence of the iterated estimand/estimator.
- This proposition shows the convergence holds for the influence function and the variance as well.

Three applications

	Jacobian γ_i	weight channel	iteration
BLP	large	active	yes
BPP	$= 0$	active	none
AJRY	modest	active	yes

- BLP: misspecification reorders *sensitivity*.
- BPP: the *choice of weight* as an efficiency–informativeness trade.
- AJRY: J and Δ *disagree*.

BLP: sensitivity reorders



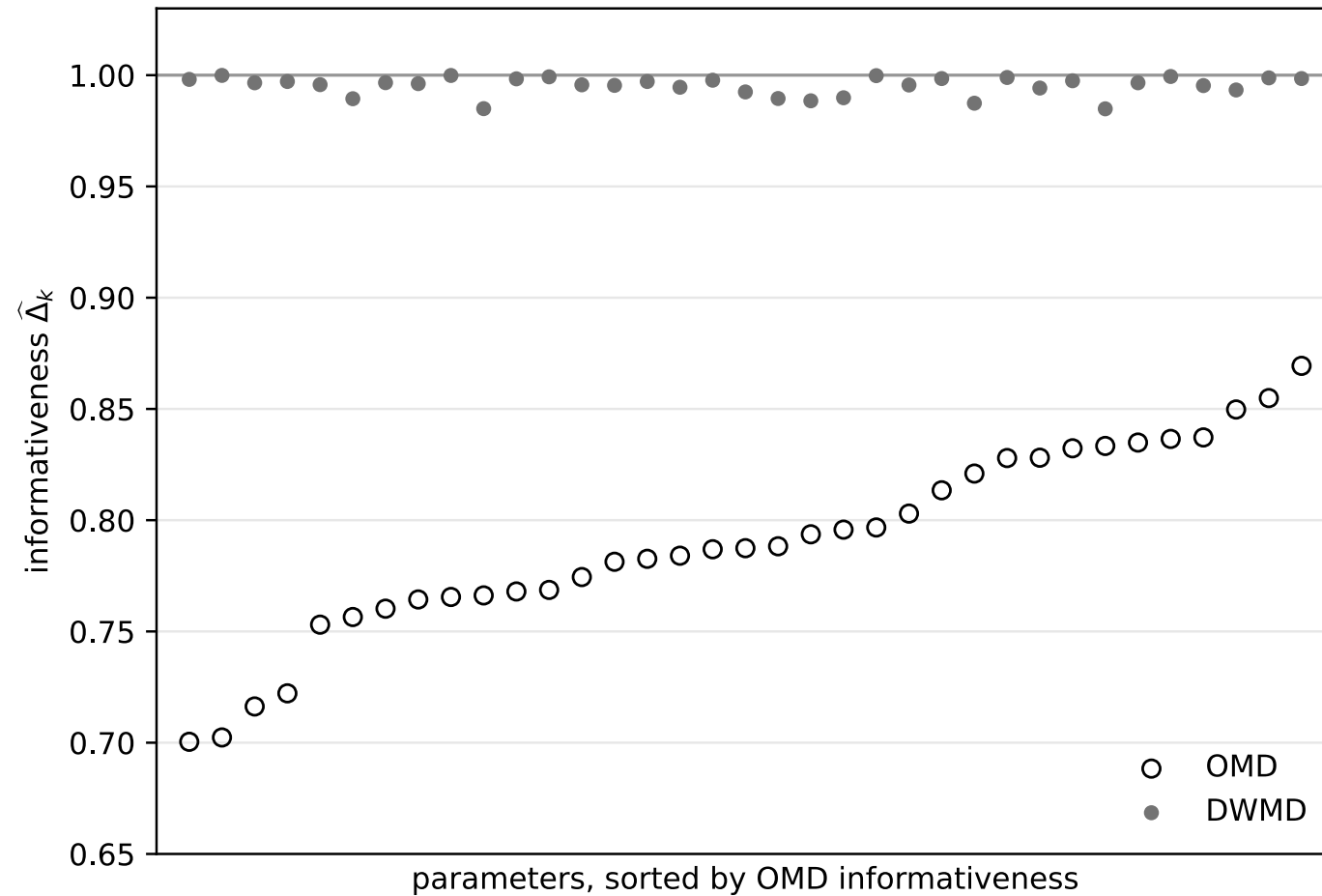
- AGS (hollow) vs. MRS (filled) for the average markup.
- $\hat{\Delta}_{markup} = 0.56, J = 345$ (p -value < 0.001).

BPP: three weights, same moments

35 parameters, $q = 325$, $n = 1,765$, $J = 538.7$ (df 290).

	$\hat{\phi}$ (permanent)	SE	$\hat{\Delta}$	$\hat{\psi}$ (transitory)	SE	$\hat{\Delta}$
OMD	0.33	0.043	0.79	0.07	0.039	0.79
DWMD	0.68	0.113	0.99	0.03	0.043	0.99
EWMD	0.61	0.109	1.00	0.00	0.049	1.00

BPP: efficiency vs. informativeness



OMD **0.70–0.87**; DWMD $\approx$ **0.99**; EWMD = **1**. $\text{SE}(\phi)$: **0.043** $\rightarrow$ **0.113**.

AJRY: income and democracy

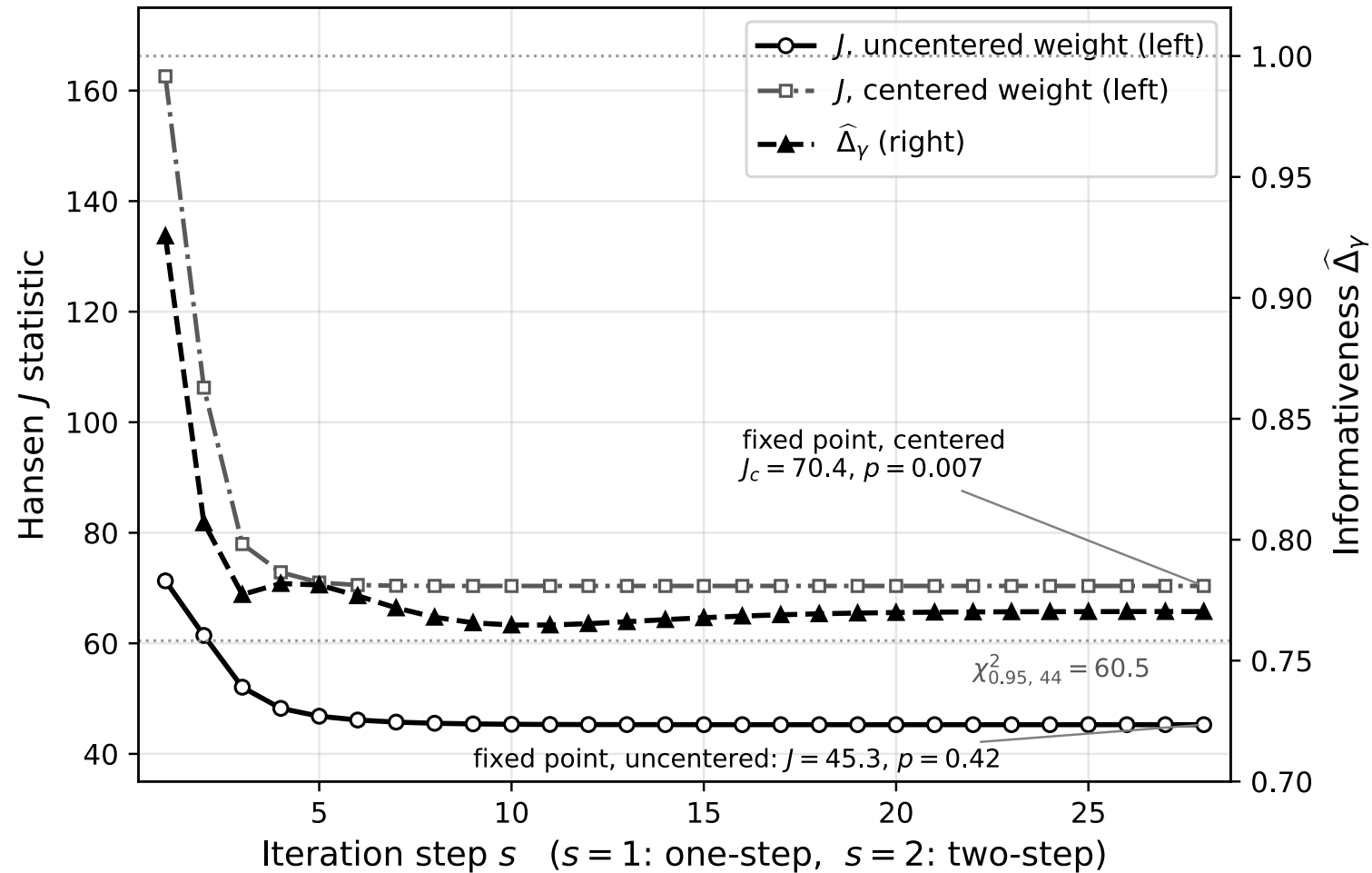
Difference GMM, $n = 127$ countries, df 44; coefficient γ .

estimator	$\hat{\gamma}$	J	p	$\hat{\Delta}_{\gamma}$
one-step (AB weight)	−0.129	71.3	0.006	0.93
two-step efficient	−0.012	61.4	0.04	0.81
iterated fixed point	−0.009	45.3	0.42	0.77

- Efficient weight lowers $\hat{\Delta}_{\gamma}$: 0.93 → 0.81 → 0.77.
- J stops rejecting (uncentered efficient weight).
- Sensitive to the choice of uncentered/centered efficient weight:

$$J = 45.3 \ (p=0.42) \ \text{vs.} \ J_c = 70.4 \ (p=0.007), \quad J = J_c / (1 + J_c/n).$$

AJRY: J turns on a centering convention; Δ does not



$$\hat{\Delta}_\gamma = 0.77, \quad J = 45.3, \quad J_c = 70.4.$$

J and Δ are complements

	J under iteration	$\hat{\Delta}$ under iteration	
BLP	flat at 345	0.56 $\rightarrow$ 0.24	J silent, Δ falls
BPP	no iteration	0.79 / 0.99 / 1	how to weight matters
AJRY	61.4 $\rightarrow$ 45.3	0.81 $\rightarrow$ 0.77	J verdict flips, Δ falls

- J : how far are the moments from zero?
- Δ : how much of the variance do the moments explain?

Takeaways

1. Robust to misspecification: evaluate sensitivity and informativeness at the pseudo-true value.
2. Sensitivity rankings can reorder (BLP).
3. Weight choice in MD estimation is an efficiency–informativeness trade (BPP).
4. Reporting $\hat{\Delta}_k$ is straightforward if you calculate misspecification-robust standard errors: it is free and answers what the J -test cannot (AJRY).

Appendix: Assumption 1 (regularity)

ASSUMPTION 1 (regularity)

- i. $\{X_i\}_{i=1}^n$ i.i.d.
- ii. $\Theta \subset \mathbb{R}^p$ compact; unique interior minimizer θ_0 of the population objective.
- iii. $g(X, \cdot)$ thrice continuously differentiable; $\mathbb{E} \sup_{\theta} \|g\|^2$, $\mathbb{E} \sup_{\theta} \|G\|^2 < \infty$, and uniform L^2 (L^1) bounds on second (third) derivatives.
- iv. curvature matrices A and $A + B$ nonsingular.
- v. (iterated GMM) the population updating map is a contraction at θ_0 ($\rho(-A^{-1}B) < 1$).
- vi. weight conditions: (a) one-step, $\widehat{W} \xrightarrow{p} W \succ 0$, $\sqrt{n} \text{vec}(\widehat{W} - W) = n^{-1/2} \sum_i \omega_i + o_p(1)$; (b) parameter-dependent, $\widehat{\phi} \xrightarrow{p} \phi_0$ asymptotically linear, $W(\phi)$ smooth with uniform convergence of $\widehat{W}(\phi)$ and its derivative.

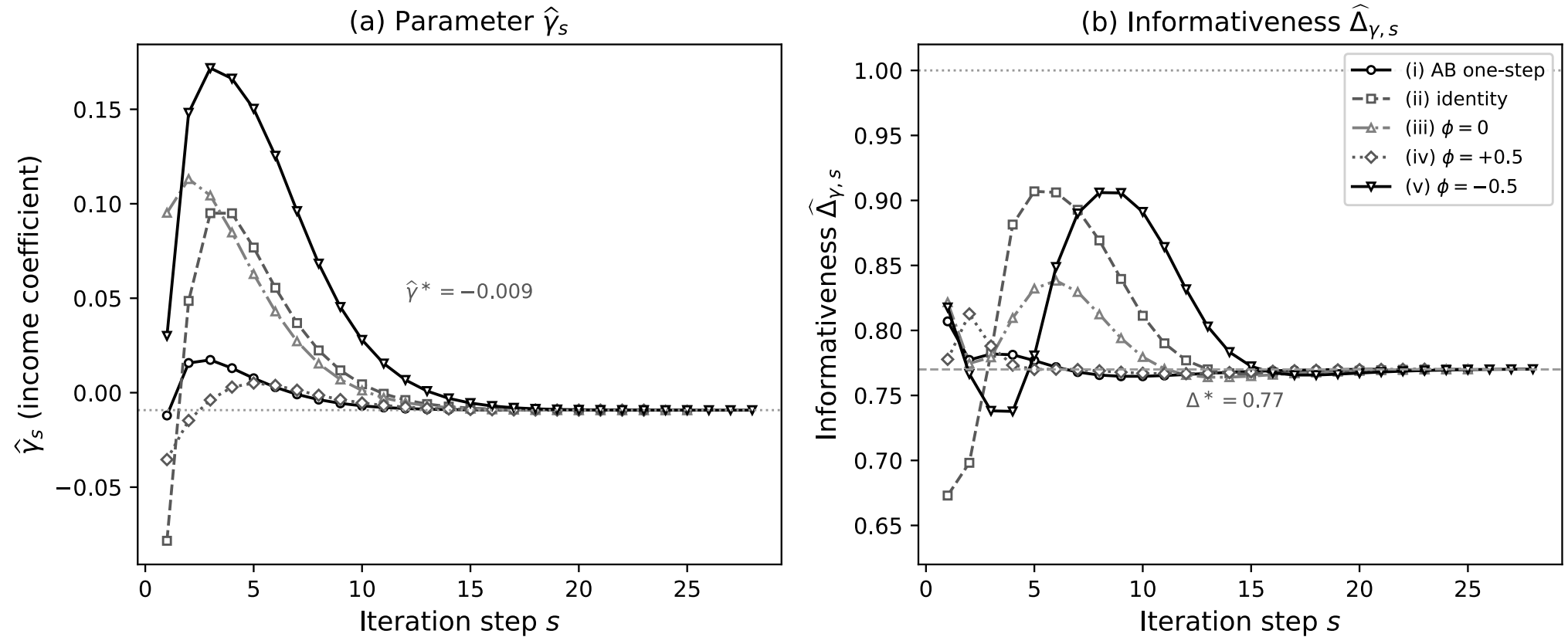
Return to:

Appendix: Notation

- Primitives: $g = \mathbb{E}[g(X_i, \theta_0)]$, $G = \mathbb{E}[\partial g(X_i, \theta_0)/\partial \theta']$, $W = \text{plim } \widehat{W}$.
- Channels: $\nu_i = g(X_i, \theta_0) - g$, $\gamma_i = \text{vec}(G(X_i, \theta_0)' - G')$, $\omega_i = \text{influence function of } \widehat{W}$, $\psi^\phi = \text{influence function of the first-step } \widehat{\phi}$.
- Curvature: $R = \mathbb{E}[\partial_{\theta'} \text{vec}(G(X_i, \theta_0)')]$, $H = (g'W \otimes I_p)R$, $A = G'WG + H$.
- Iteration: $S = \partial_{\phi'} \text{vec } W(\phi_0)$ (two-step) or $\partial_{\theta'} \text{vec } W(\theta_0)$ (iterated), $B = (g' \otimes G')S$.
- Coefficient maps: $M_\nu = -A^{-1}G'W$, $M_\gamma = -A^{-1}(g'W \otimes I_p)$, $M_\omega = -A^{-1}(g' \otimes G')$.

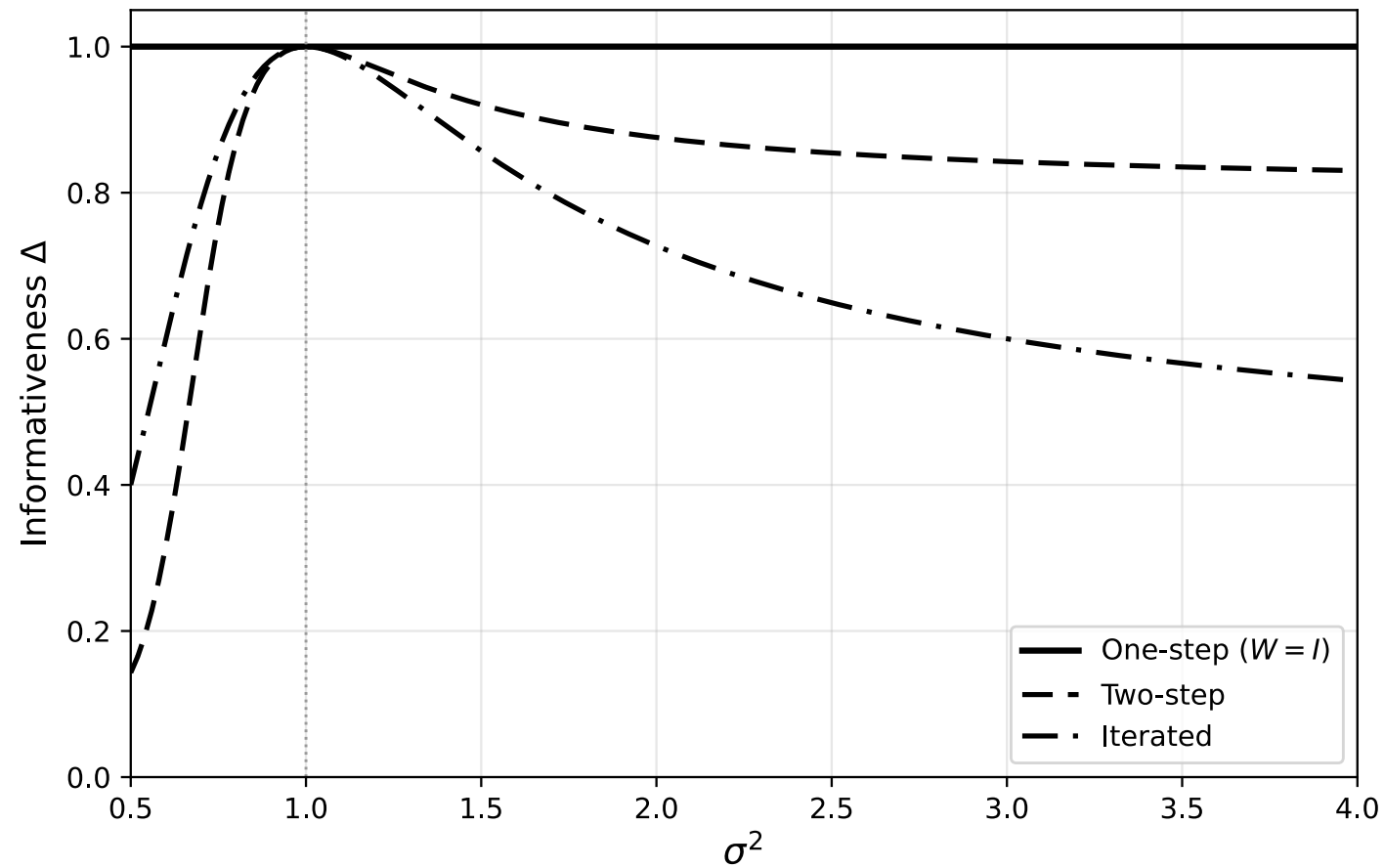
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Appendix: AJRY multistart, common fixed point



$\hat{\gamma}_s$ and $\hat{\Delta}_{\gamma,s}$ reach the common fixed point from every start; the approach of Δ is non-monotone.

Appendix: Schennach closed form



Schennach (2007): normal mean with misspecified variance restriction; closed-form Δ for one-step, two-step, iterated, CUE.

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